

Predicting Molecular Energy Gaps using Graph Neural Networks

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Abstract—The HOMO-LUMO gap is a key molecular property used in materials and drug discovery, but computing it through Density Functional Theory (DFT) is slow and expensive. We train Graph Neural Networks (GNNs) to predict this gap directly from 2D molecular graph structure on the PCQM4Mv2 dataset (3.4M training molecules). Specifically, we compare a 7-layer GINE backbone against the same backbone augmented with a Virtual Node, a single global node connected to every atom and intended to address the over-squashing bottleneck on larger molecules. After a hyperparameter search on a 200k subset to fix architecture and learning rate, both final models were trained for five epochs on the full training set on an H100 GPU. The Baseline GINE reached a validation MAE of 0.1421 eV, and the Virtual Node GINE reached 0.1487 eV, so the Virtual Node did not improve aggregate validation MAE. However, a size-bucket analysis showed that the Virtual Node reduced MAE on molecules with 21-40 atoms, where long-range information flow matters more. The takeaway is that global communication mechanisms can help in specific regimes, but are not automatically beneficial when the baseline is already deep and edge-aware.

I. INTRODUCTION

The HOMO-LUMO gap of a molecule is the energy difference between its highest occupied and lowest unoccupied molecular orbitals. It is one of the simplest scalar descriptors of chemical reactivity, and it shows up in nearly every materials and drug discovery pipeline as a screening or filtering criterion. The standard way to compute it is Density Functional Theory (DFT), a quantum-mechanical simulation that solves an approximation of the electronic Schrödinger equation for the molecule. DFT is accurate, but it can take hours of compute per molecule. This is fine for one-off calculations, but unworkable when screening millions of candidate compounds, which is exactly the scale at which modern discovery operates.

The motivation for using machine learning here is straightforward. If a model can take a molecular graph as input and predict the HOMO-LUMO gap in milliseconds, the cost-per-molecule drops by several orders of magnitude. Researchers can then prioritize which candidates to actually run through DFT, or skip DFT entirely once accuracy is high enough. Faster screening means faster discovery of better solar cells, batteries, catalysts, and drug candidates.

We use the PCQM4Mv2 dataset from the OGB Large-Scale Challenge [3], which contains 3,378,606 training and 73,545 validation molecules with DFT-computed HOMO-LUMO gaps. Each molecule is provided as a SMILES string and converted to a graph using RDKit, where atoms become

nodes with 9 categorical features (atomic number, chirality, degree, formal charge, etc.) and bonds become edges with 3 categorical features (bond type, stereo, conjugation). The dataset also provides 3D conformer coordinates, but we use only the 2D molecular graph in this work.

Our approach was a controlled comparison between a deep, edge-aware message-passing GNN and the same backbone augmented with a Virtual Node, which is a single learned node connected to every atom in the molecule. The motivation is the over-squashing bottleneck identified by Alon and Yahav [2]: in standard message passing, information from distant atoms must be compressed through bottleneck paths, which limits how well the model can predict global properties of large graphs. A Virtual Node provides a constant-hop shortcut between any two atoms.

We anticipated compute and convergence to be the dominant constraints. PCQM4Mv2 is large enough that one full training pass takes several hours on a high-end GPU, so we planned the final experiments around 5-epoch runs. The first thing we tried, a 5-layer GINE with hidden dimension 256, was reasonable but not the best in our hyperparameter search; adding two more message-passing layers improved subset validation MAE more than other changes we considered. The final comparison therefore used 7-layer GINE backbones with and without the Virtual Node.

II. RELATED WORK

A. Molecular Property Prediction

The OGB Large-Scale Challenge [3] established a standardized large-scale evaluation suite for molecular property prediction, including the PCQM4Mv2 dataset that we use here. By fixing the training, validation, and test splits across submissions, the benchmark allows fair architectural comparisons at a scale that smaller datasets such as QM9 cannot reach.

B. Message Passing Neural Networks

Gilmer et al. [1] introduced Message Passing Neural Networks (MPNN) as a unifying view of graph neural network architectures applied to chemistry. Each MPNN layer consists of a message-passing phase, where each node aggregates messages from its neighbors, and an update phase, where each node updates its hidden state from the aggregated message. Most modern molecular GNN architectures, including the GINE backbone we use, fit cleanly under this framework.

C. GIN and GINE Models

Xu et al. [4] introduced the Graph Isomorphism Network (GIN) and showed that its sum-aggregation update rule yields expressive power equal to the Weisfeiler-Lehman graph isomorphism test, which is the strongest any message-passing GNN can be. GINE is a small extension that incorporates edge features directly into the message function. This matters for molecular graphs, since bond types (single, double, aromatic, etc.) carry chemically relevant information that a node-only model would discard. Our backbone is GINE.

D. Long-Range Communication and Virtual Nodes

Alon and Yahav [2] formalized the over-squashing problem in message-passing GNNs: when information must traverse a long path through a graph, it gets exponentially compressed at bottlenecks, so global properties of large graphs become difficult to learn. Two main families of solutions exist. The first is the Virtual Node, originally proposed as the “master node” in the MPNN paper [1], which is a single node added to the graph that connects to every other node and provides a constant-hop global communication channel. The second is graph transformers such as Graphormer [5], which use full self-attention between all nodes to enable global communication at the cost of significantly higher parameter count and compute. We chose the Virtual Node approach because it adds relatively few parameters and lets us isolate the effect of global communication on top of an otherwise standard GINE backbone, rather than confounding it with attention or a much larger model.

III. DATASET AND REPRESENTATION

A. PCQM4Mv2 Dataset

PCQM4Mv2 contains 3,378,606 training molecules and 73,545 validation molecules drawn from PubChemQC, with HOMO-LUMO gaps labeled in eV by DFT calculation at the B3LYP/6-31G* level. Each molecule is provided as a SMILES string and converted to a graph using RDKit. Atoms become nodes with 9 categorical features (atomic number, chirality, degree, formal charge, number of hydrogens, hybridization, aromaticity, ring membership, and radical electrons), and bonds become edges with 3 categorical features (bond type, stereo, and conjugation). Categorical features are mapped to dense vectors via learned embeddings inside the model.

The label distribution is concentrated between roughly 2 and 10 eV, with a long tail on both ends. Molecule sizes range from a small number of atoms to roughly 50, with most molecules falling in the 10–30 atom range. The dataset also provides 3D conformer coordinates, but we did not use them; this work studies what is achievable from 2D graph connectivity alone.

IV. METHODS

We treat HOMO-LUMO gap prediction as a regression task: given a molecular graph, output a single scalar in eV. Both of our final models share the same overall architecture. Atom and bond categorical features are embedded by summing learned per-feature embeddings, which gives a hidden vector

of dimension 256 per atom and per bond. The resulting node and edge representations are passed through $L = 7$ stacked GINE layers. Each layer applies a GINE convolution, batch normalization, ReLU, dropout, and a residual connection back to the layer input. Per-layer atom embeddings are pooled to graph level by mean pooling, and the pooled representations from all layers are summed to form the final graph embedding, a simple form of jumping knowledge that lets the prediction head use information from multiple message-passing depths. A small two-layer MLP maps the graph embedding to the predicted gap.

The GINE update rule is:

$$h_v^{(l+1)} = \text{MLP}^{(l)} \left((1 + \epsilon^{(l)}) h_v^{(l)} + \sum_{u \in \mathcal{N}(v)} \text{ReLU}(h_u^{(l)} + e_{vu}^{(l)}) \right) \quad (1)$$

where $h_v^{(l)}$ is the embedding of node v at layer l , $e_{vu}^{(l)}$ is the embedding of the bond between u and v at layer l , $\mathcal{N}(v)$ is the set of neighbors of v , and $\epsilon^{(l)}$ is a learnable scalar that controls how much a node’s own embedding is preserved relative to the aggregated neighbor messages.

Our implementation uses PyTorch and PyTorch Geometric [6], with molecular graph construction handled by RDKit [7] via the OGB smiles2graph utility. The model code, training loop, and evaluation script were written from scratch; we reused only the dataset loader and the official PCQM4Mv2Evaluator from the OGB package.

A. Virtual Node Variant

The Virtual Node variant adds a single global node g to each molecular graph. At the start of every forward pass, the Virtual Node embedding is initialized from a learned vector $h_g^{(0)}$ shared across all molecules in a batch. Before each GINE layer, the current Virtual Node embedding is broadcast to every atom in the molecule and added to its embedding:

$$\tilde{h}_v^{(l)} = h_v^{(l)} + h_g^{(l)}. \quad (2)$$

The GINE layer then runs on these Virtual-Node-augmented atom embeddings. Afterwards, the Virtual Node itself is updated by summing the post-layer atom embeddings inside the molecule, adding the previous Virtual Node state, and passing the result through a two-layer MLP:

$$h_g^{(l+1)} = \text{MLP}_g^{(l)} \left(h_g^{(l)} + \sum_{v \in V} h_v^{(l+1)} \right). \quad (3)$$

This gives every atom a constant-hop pathway to every other atom in the same molecule: information from an atom can flow into the Virtual Node in one update step, and back out to any other atom on the next layer. The added cost over the baseline is one extra MLP per layer plus the broadcast addition, which is small relative to the cost of the GINE layers themselves.

B. Hyperparameter Search

First, we completed a hyperparameter search on a subset of the molecules. We used 200,000 for training and 20,000 for

validation, allowing us to compare architecture and optimization choices without training on the full dataset each time. We varied the hidden dimension, number of message-passing layers, learning rate, and also included a run with the Virtual Node.

Our best result came from increasing the number of message-passing layers, while reducing the hidden dimension hurt performance. This suggests that the smaller model lacked sufficient representational capacity. The reduction in learning rate also performed worse within the same 5-epoch training setup. The initial Virtual Node test for our hyperparameter search did not outperform the 7-layer baseline, so for the final comparison we used matched 7-layer models with and without the Virtual Node.

C. Final Full-Dataset Training

For the final experiments, we trained two models on the full PCQM4Mv2 training split of 3,378,606 molecules:

- **Baseline GINE:** 7 GINE layers, hidden dimension 256, no Virtual Node.
- **Virtual Node GINE:** same 7-layer GINE backbone, with Virtual Node enabled.

Both models used batch size 256, learning rate 3×10^{-4} , dropout 0.1, and the Adam optimizer [8]. Training minimized mean absolute error (L1 loss) directly, which matches the official PCQM4Mv2 evaluation metric. Due to the heavy computation requirements, each model was trained for five full epochs on an H100 GPU on a cloud computing platform, which took about 3 hours per run. This was why we had to limit the number of epochs to 5. The best checkpoint was selected by validation MAE on the official 73,545-molecule validation split.

V. RESULTS

A. Final Validation MAE

Both final models were trained for five epochs on the full training set. For the evaluation, we report our best validation checkpoint. The Baseline GINE model reached its best validation MAE at epoch 5, while the Virtual Node model did so at epoch 3.

Table II reports the final validation performance on the official PCQM4Mv2 validation split. For the Baseline GINE model, the validation MAE was 0.1421 eV, while for the Virtual Node model, it was 0.1487 eV. Based on this, the Virtual Node did not improve aggregate validation MAE over the GINE baseline, which actually slightly outperformed the Virtual Node model.

The Baseline GINE model outperformed the Virtual Node model by 0.0066 eV. This shows that adding a global communication node is not automatically helpful if the baseline is sufficiently complex. Since both final models were evaluated from one training run, the difference should be interpreted as suggestive rather than statistically conclusive. However, the result does support the need for careful baseline comparison rather than assuming that a more complex architecture will necessarily improve performance.

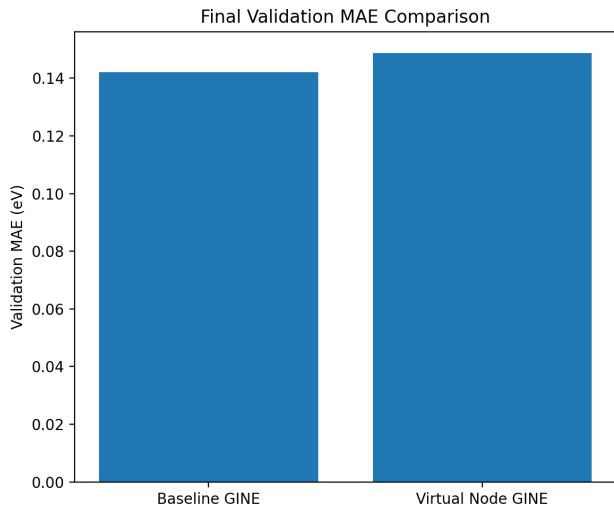


Fig. 1: Final validation MAE comparison between the Baseline GINE and Virtual Node GINE models on the official validation split. Lower is better.

Neither model showed clear signs of overfitting within the five-epoch budget. The Baseline GINE was still improving on validation at epoch 5, and the Virtual Node model peaked at epoch 3 with a small increase in validation MAE thereafter. Given that a single epoch already corresponds to 3.4M molecules, the implicit regularization from this scale, combined with dropout and batch normalization, kept both models closer to an underfitting regime than an overfitting one.

B. Parity Plots

The purpose of the parity plots was to compare the predicted HOMO-LUMO gaps against the actual DFT-computed values. Points near the diagonal correspond to accurate predictions. Further deviations from the diagonal imply larger prediction error. Figure 2 shows parity plots for both final models.

Both models show a strong positive relationship between the predicted and real HOMO-LUMO gaps. This means our learned graph representation is accurate and can capture meaningful structural information. However, neither model produces a perfect diagonal alignment. Some molecules remain difficult, as shown by the spread around the diagonal. There are many outlier cases with large prediction error. We therefore cannot claim that the model has fully learned the underlying quantum chemistry. What we can say from this parity plot is that GNNs learn useful predictive patterns from molecular graph structure.

C. Error Distribution

Figure 3 shows the distribution of absolute prediction errors on the validation set for both models.

Most of the validation molecules have small absolute errors. A smaller number of molecules produce much larger errors. This means that overall MAE is influenced by a long tail of

TABLE I: Hyperparameter search on a 200k/20k train-validation subset. The 7-layer GINE model achieved the best subset validation MAE and was selected for full-dataset training.

Run	Hidden Dim	Layers	Learning Rate	Virtual Node	Parameters	Best Subset Val MAE (eV)
Base GINE	256	5	3×10^{-4}	No	1.398M	0.2302
Narrow GINE	128	5	3×10^{-4}	No	0.363M	0.2402
Deeper GINE	256	7	3×10^{-4}	No	1.925M	0.2171
Low-LR GINE	256	5	1×10^{-4}	No	1.398M	0.2413
VN Pilot	256	5	3×10^{-4}	Yes	2.526M	0.2347

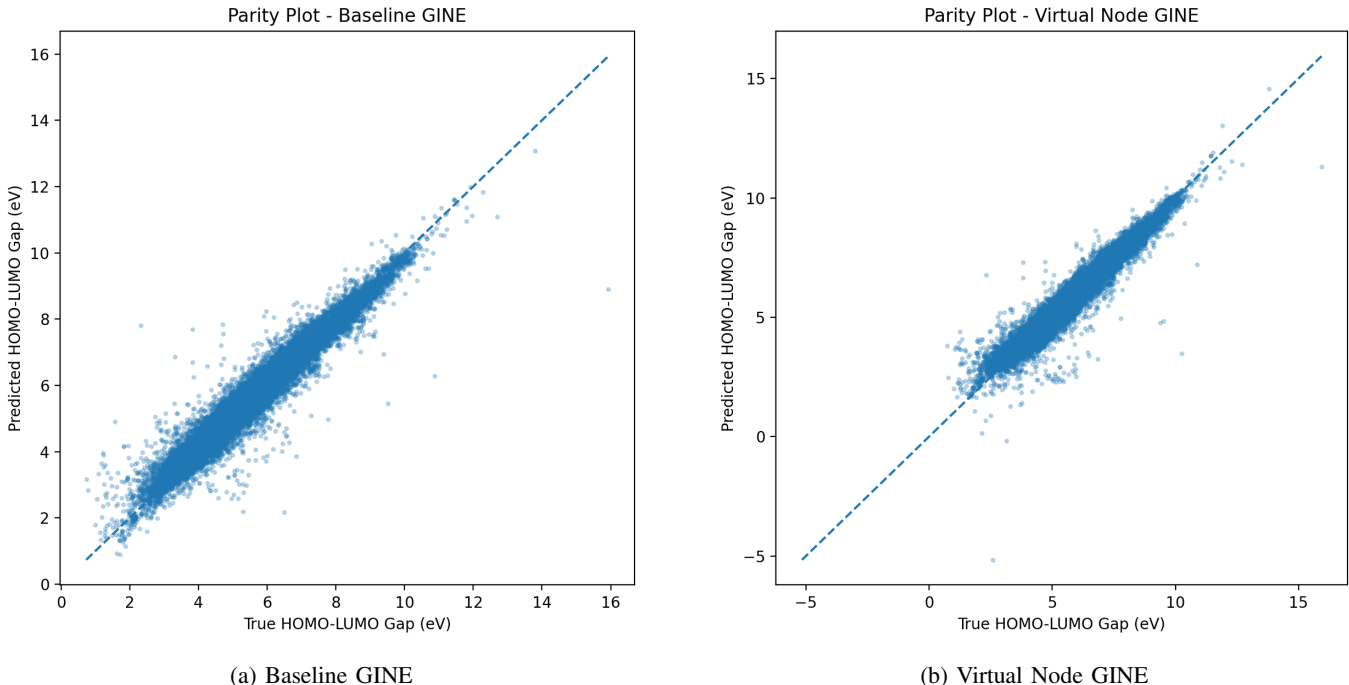


Fig. 2: Parity plots for final validation predictions. The dashed diagonal represents perfect prediction.

TABLE II: Final validation performance on the official PCQM4Mv2 validation split. Both models were trained for five epochs, and the reported MAE corresponds to the best validation checkpoint saved during training.

Model	Best Checkpoint Epoch	Validation MAE (eV)
Baseline GINE	5	0.1421
Virtual Node GINE	3	0.1487

harder molecular examples. The distributions for both models are similar, which corresponds with the small aggregate MAE difference between the Baseline GINE and the Virtual Node GINE.

D. Molecule Size Analysis

Our original motivation for the Virtual Node was to improve long-range communication. As a result, we evaluated validation MAE as a function of molecule size. We split the validation molecules into buckets based on the number of atoms and computed the average absolute error for each bucket. This allowed us to test whether the Virtual Node

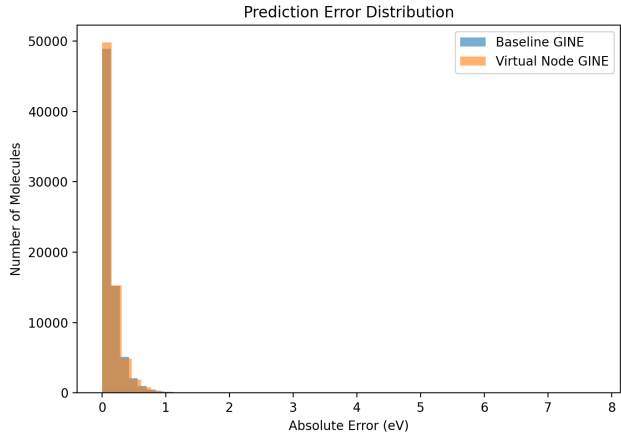


Fig. 3: Distribution of absolute prediction errors for the Baseline GINE and Virtual Node GINE models.

helped with larger molecular graphs, where it is more difficult to transmit information across the entire molecule.

Table III reports the numerical MAE values by molecule

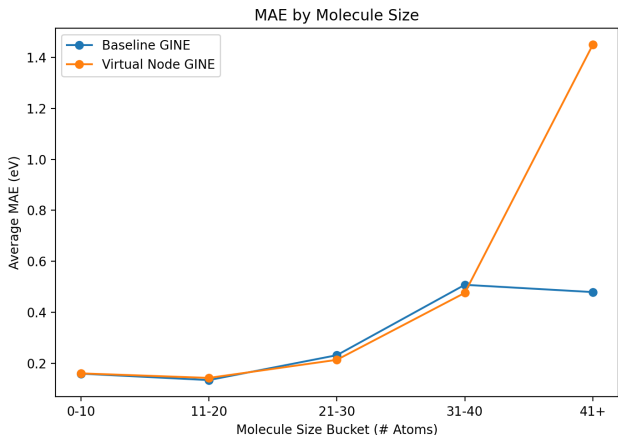


Fig. 4: Validation MAE by molecule size bucket. This analysis evaluates whether the Virtual Node improves performance more strongly on larger molecular graphs.

size bucket. The Baseline GINE performs slightly better on the smallest molecule buckets, while the Virtual Node performs better on the 21–30 and 31–40 atom buckets. This is a more nuanced result than the aggregate validation MAE alone. Although the Virtual Node did not improve overall validation MAE, it did reduce error for some medium-to-large molecule buckets, which partially supports the hypothesis that global communication can help when molecules become larger.

TABLE III: Validation MAE by molecule size bucket. Lower is better.

Size Bucket	Baseline GINE MAE	VN GINE MAE
0–10	0.1590	0.1601
11–20	0.1337	0.1422
21–30	0.2314	0.2133
31–40	0.5078	0.4761

The size-bucket analysis gave us insight that the aggregate MAE alone could not provide. As shown in the table, the MAE is better for the Virtual Node GINE for the larger size buckets. The Virtual Node appears to help for larger molecules where long-range information flow is more important. A stronger conclusion would require multiple random seeds and a more careful analysis of how many molecules fall into each size bucket.

VI. DISCUSSION

Our experiments addressed the main objectives from the proposal: predicting the HOMO-LUMO gaps from molecular graphs. We were able to compare a standard message-passing GNN against a Virtual Node variant and evaluate whether or not the Virtual Node helped on larger molecular graphs. Our first objective was partially met. While our best model achieved a validation MAE of 0.1421, it did not reach our original target of 0.12. Regardless, we were able to show that the model learned a meaningful graph-to-property mapping from molecular structure.

We did not reach our original target for a few reasons. First, the final models were only trained for five epochs due to compute limits. The dataset is large, and more passes may have been able to yield better results. Second, our architecture was lightweight, as we used a GINE-style model rather than a more complex graph transformer, such as one that could have included attention mechanisms. Also, our model only used the 2D molecular graph connectivity, and HOMO-LUMO gaps depend on spatial and electronic structure as well, so the lack of 3D geometry harmed our predictive accuracy. These all serve as limitations for our project. Future work should include longer training, multiple seeds, stronger Virtual Node update functions, and comparison against more global architectures such as graph transformers.

The second objective was met experimentally through a controlled comparison between matched Baseline GINE and Virtual Node GINE models. Both final models used the same hidden dimension, number of layers, learning rate, dropout, batch size, and training split. Contrary to our initial hypothesis, the Virtual Node did not improve aggregate validation MAE: the Baseline GINE achieved 0.1421 eV, while the Virtual Node GINE achieved 0.1487 eV. This suggests that adding a global communication pathway is not automatically beneficial when the baseline is already deep, edge-aware, and trained on a large dataset.

Our molecule-size analysis, on the other hand, provided a more nuanced interpretation and useful result, one that we actually did not set out to discover. We noticed the Virtual Node performed worse on smaller molecules but improved MAE as they got larger. This partially supported our original motivation that global communication can help improve MAE, mainly reinforcing the idea that molecular graphs become more difficult when they are larger and long-range information flow becomes more important. Our final conclusion for this project therefore is mixed: the Virtual Node was not the best model overall, but it does help for the purpose that it was designed for.

VII. CONCLUSION

We trained two GINE-based GNNs on the full PCQM4Mv2 training set to predict molecular HOMO-LUMO gaps. Our matched comparison between a 7-layer Baseline GINE and the same backbone augmented with a Virtual Node showed that, at this scale and depth, the Virtual Node did not improve aggregate validation MAE: 0.1421 eV for the baseline against 0.1487 eV for the Virtual Node variant. The size-bucket analysis was the more informative result. The Virtual Node reduced MAE on molecules with 21–40 atoms, where long-range information flow becomes a real bottleneck, but performed slightly worse on smaller molecules where the baseline already had short paths between all atoms. This is consistent with the over-squashing motivation, but not with the simpler claim that a Virtual Node is always helpful. With longer training, multiple random seeds, 3D coordinate features, or a stronger global mechanism such as graph attention, we

would expect both models to improve and the regime in which the Virtual Node helps to become more clearly defined.

TEAM CONTRIBUTIONS

TABLE IV: Team contributions.

Member	Contributions
Kapil Meena	Infrastructure setup, experiment execution support, report writing.
Gaurav Mitra	Custom GNN implementation, training loop, report writing.
Archith K. Iyer	Hyperparameter tuning, full-dataset training setup, final evaluation script, validation metrics, result visualizations, and report restructuring.

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